DOD-STD-1312-111
26 March 1984

SUPERSEDING
DOD-STD-1312-11
19 October 1983

MILITARY STANDARD

FASTENER TEST METHODS, METRIC

METHOD 111

TENSION FATIGUE

FSC 53GP

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Fastener Test Methods, Metric, Method 111, Tension Fatigue

DOD-STD-1312-111

1. This Military Standard is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specification and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

This standard sets forth standard test procedures for determining tension-fatigue of bolts, screws and nuts at room temperature.

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1. SCOPE

1.1 Applicability. This test method establishes the procedure and equipment to be used for tension-fatigue testing of bolts, screws and nuts at room temperature.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the current Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto (if applicable), form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels (Aerospace Practice) Process for
MIL-C-15074	Corrosion Preventive, Fingerprint Remover

STANDARDS

FEDERAL

FED-STD-H28/21	Screw Thread Standards for Federal Services Section 21, Metric Screw Threads

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	Surface Texture (Surface Roughness, Waviness, and Lay)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus.

4.1.1 Fatigue test machine.

4.1.1.1 Load measuring system. The dynamic load measuring system shall be accurate within ±2 percent of the applied maximum load.

4.1.1.2 Load maintaining system. The load maintaining system shall be capable of maintaining the required load. Machines operating without load maintainers shall be monitored at intervals by the operator. The intervals shall be determined according to the expected cycle life and equipment capability.

4.1.1.3 Calibration. The load accuracy of each machine shall be verified after each 1000 hours of operating time or every six months, whichever occurs first, and after any machine modification or relocation which could affect the load accuracy. Methods of calibration and alignment shall be as specified in equipment appendix of MIL-STD-1312.

4.1.2 Fixtures. The fixtures used for fatigue testing fasteners shall be designed to incorporate all of the features outlined in figure 1.

4.1.2.1 Spacer. A spacer conforming to figure 1 may be used under the head of protruding-head fasteners to provide clearance for the fastener head-shank fillet radius. Plain flat spacers may be used under the nut.

4.1.2.2 Test nuts. The fatigue test nuts shall conform to figure 2 or 3. Threaded adapters may be used on shear fasteners through 12 millimeter diameter or tension fasteners too short to test with nuts on available fixtures. Locking devices shall not be permitted. Type A or B nuts shall be used for testing bolts under 1800MPa.

4.1.2.2.1 Reused test nuts. A fatigue test nut may be reused for a maximum of five tests, providing the nut meets the dimensional requirements and has not been damaged by a previous test. The same nut shall be used for testing the same lot of bolts. For referee tests, only unused nuts shall be used.

4.1.2.3 Test bolts. Test bolts of studs shall not be reused.

4.1.2.4 Limitations. Universal joints or spherical seats shall not be used in the test column.

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4.1.2.5 Fixture alignment. The fixtures shall be capable of holding the fasteners in conformance with the alignment requirements specified in equipment appendix of MIL-STD-1312.

4.1.2.6 Threaded adapters. The hardness of threaded adapters shall conform to requirements of figure 2 test nuts.

4.2 Specimens.

4.2.1 Testing of bolts.

4.2.1.1 Protruding and flush head bolts. Protruding-head tensile bolts shall be tested using the fatigue test nuts as shown in figure 2 or threaded adapter. Flush-head bolts and protruding-head shear bolts shall be tested using the fatigue test nuts (figure 2 and 3) or threaded adapters. Threaded adapters shall be replaced when they become damaged as determined by visual inspection or when they are no longer within specified tolerance limits. Only new adapters shall be used for referee tests.

4.2.1.2 Coarse or special thread bolts. When testing bolts having coarse or special threads, the fatigue test nut shall be as specified in the applicable product specification.

4.2.1.3 Grip length. Unless otherwise specified, the minimum grip length for test bolts shall be twice the nominal diameter.

4.2.2 Testing of nuts.

4.2.2.1 Sample of bolts or studs. A sample of each lot of bolts or studs used for testing nuts shall first be tested in accordance with this standard, using test nuts in figure 2 to determine their reliability.

4.2.2.2 Nut and bolt assembly. When testing nuts, the nut shall be assembled on the bolt with not less than two full threads including the chamfer exposed between the nut and the end of the bolt.

4.2.2.3 Nut testing. Nuts shall be tested on fatigue bolts specified in the applicable procurement document.

4.2.3 Test results. Fracture in the threads of the bolt or nut and failure to support the test load shall constitute a valid test conclusion. Fracture of the bolt in any other location shall not be considered a valid test conclusion and the test shall be repeated using a new nut and bolt.

5. DETAIL REQUIREMENTS

5.1 Test procedures.

5.1.1 Bolts.

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5.1.1.1 Tension bolts. Tension bolts shall be assembled so that a minimum of two and a maximum of three threads are exposed between the nut bearing face and bolt thread run out.

5.1.1.2 Shear bolts. Shear bolts shall be assembled with a minimum of one-half and a maximum of one thread exposed between the nut bearing face and the bolt thread run out.

5.1.1.3 Flush-head bolts. Flush-head bolts shall be checked for uniform bearing between the bolt head and the loading cup in which it is to be tested. There shall be no contact in the area of the head-shank fillet radius.

5.1.1.4 Fatigue tests. Fatigue tests are not applicable to parts with cross-drilled threads.

5.2 Test conditions.

5.2.1 Torque. There shall be no tensile stress in the fasteners due to torque or other installation procedures.

5.2.2 Load level. The fatigue test load shall be as specified in the applicable procurement document or specification.

5.2.3 Load ratio. Unless otherwise specified, the low load shall be one-tenth of the high load level (R = 0.1).

5.2.4 Test speed. The test speed shall not exceed 210 Hz. If excessive heat is generated when testing high strength fasteners at high frequencies, the test speed shall be reduced to insure that the maximum (test fasteners) temperature does not exceed 66°C.

6. NOTES

6.1 Test reports. The test report shall contain the following:

a. Fastener description.

b. Part number and fastener diameter.

c. Lot identification.

d. Material.

e. Heat treat.

f. Grip length.

g. Mating part.

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h. Model and serial number of test machine.

i. Calibration date of test machine.

j. Test load.

k. Test speed.

l. Test duration (no. of cycles).

m. Type of failure (if any).

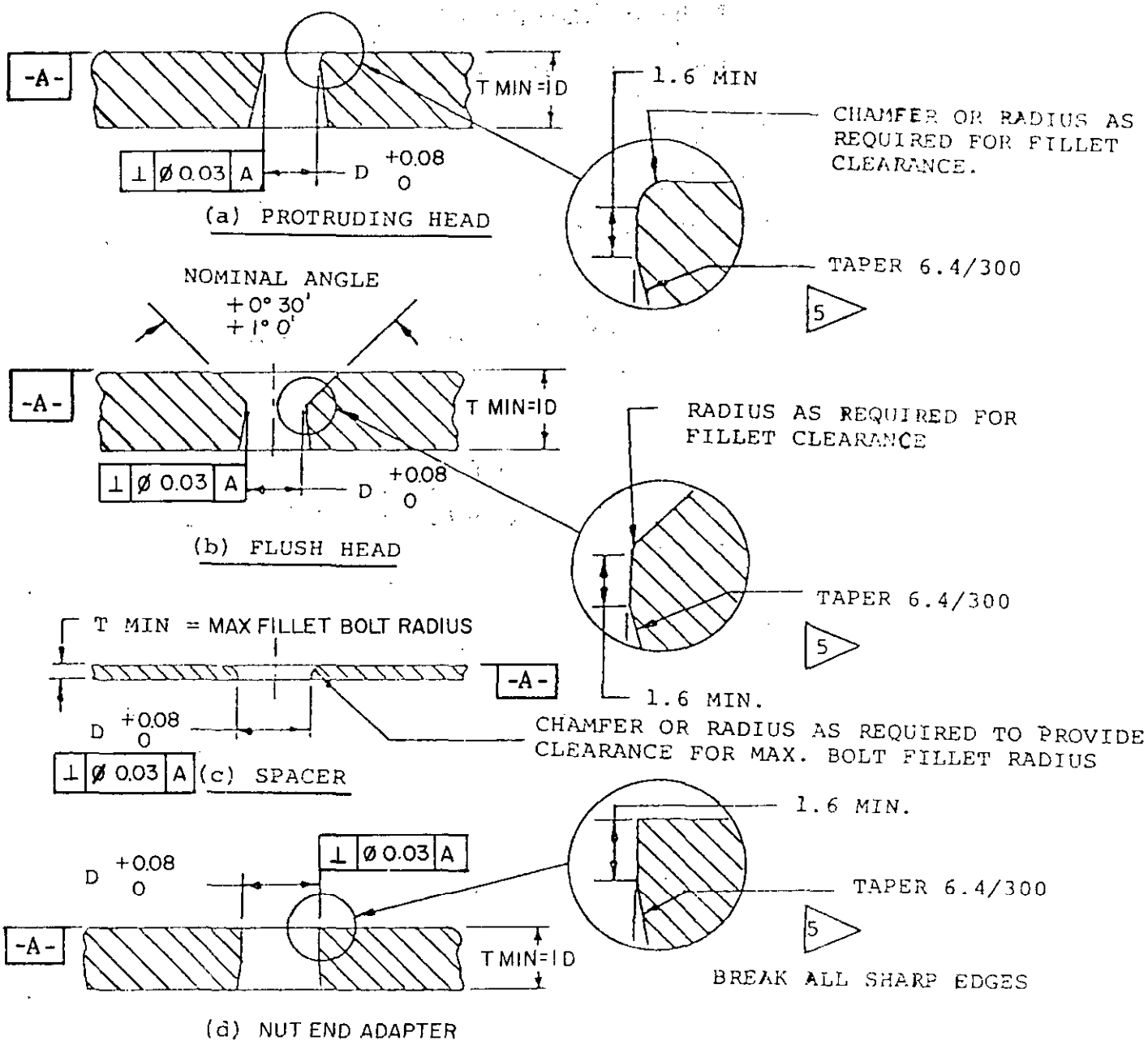
n. Cause and length of interruption of test.

Custodians:
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Preparing Activity
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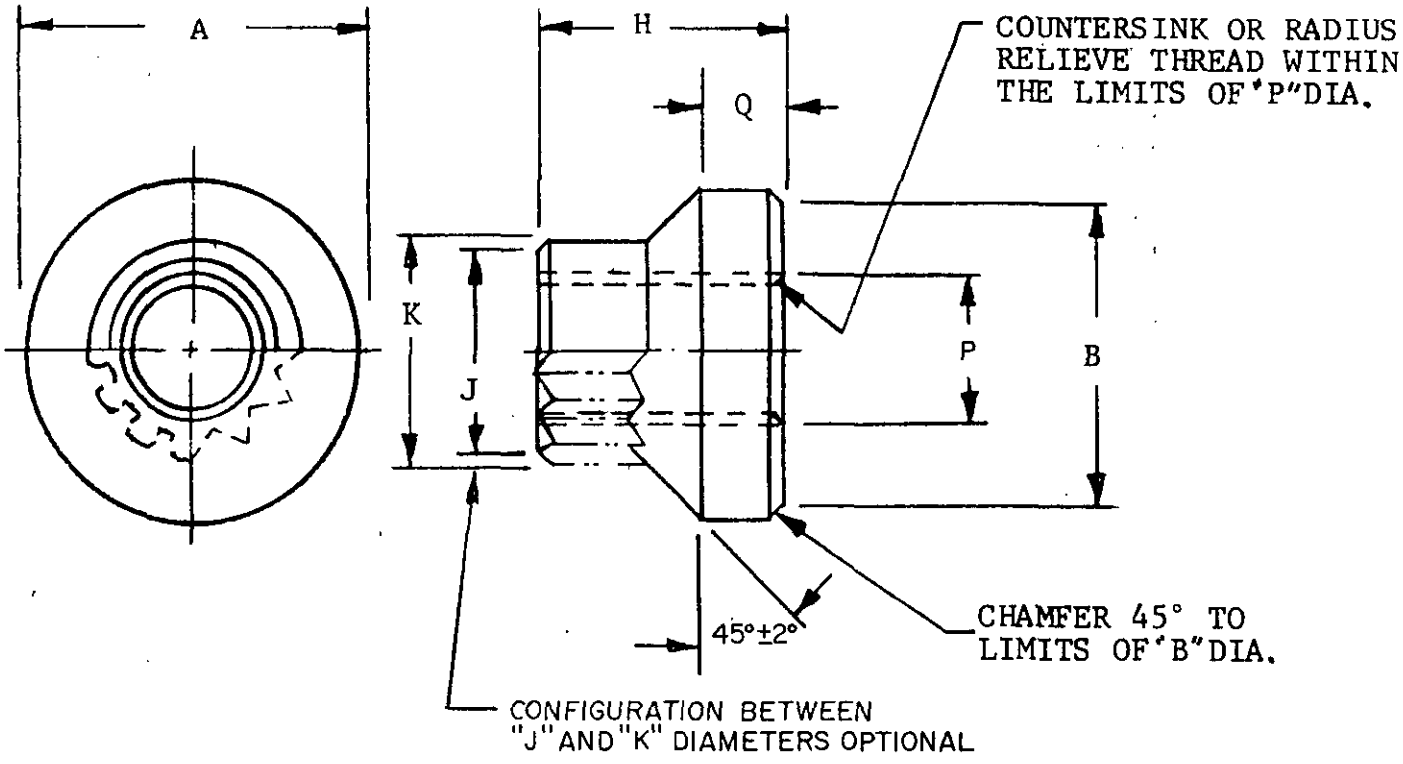


NOTES:

1. Material: Steel, Rc 43 minimum.
2. Clean as required to maintain 0.8µm finish and required tolerances on load bearing surfaces.
3. All diameters to be concentric within 0.03 FIM.
4. Unless otherwise specified, dimensions in millimeters.
5. Taper optional.
6. D is nominal diameter.

FIGURE 1. Loading cups.

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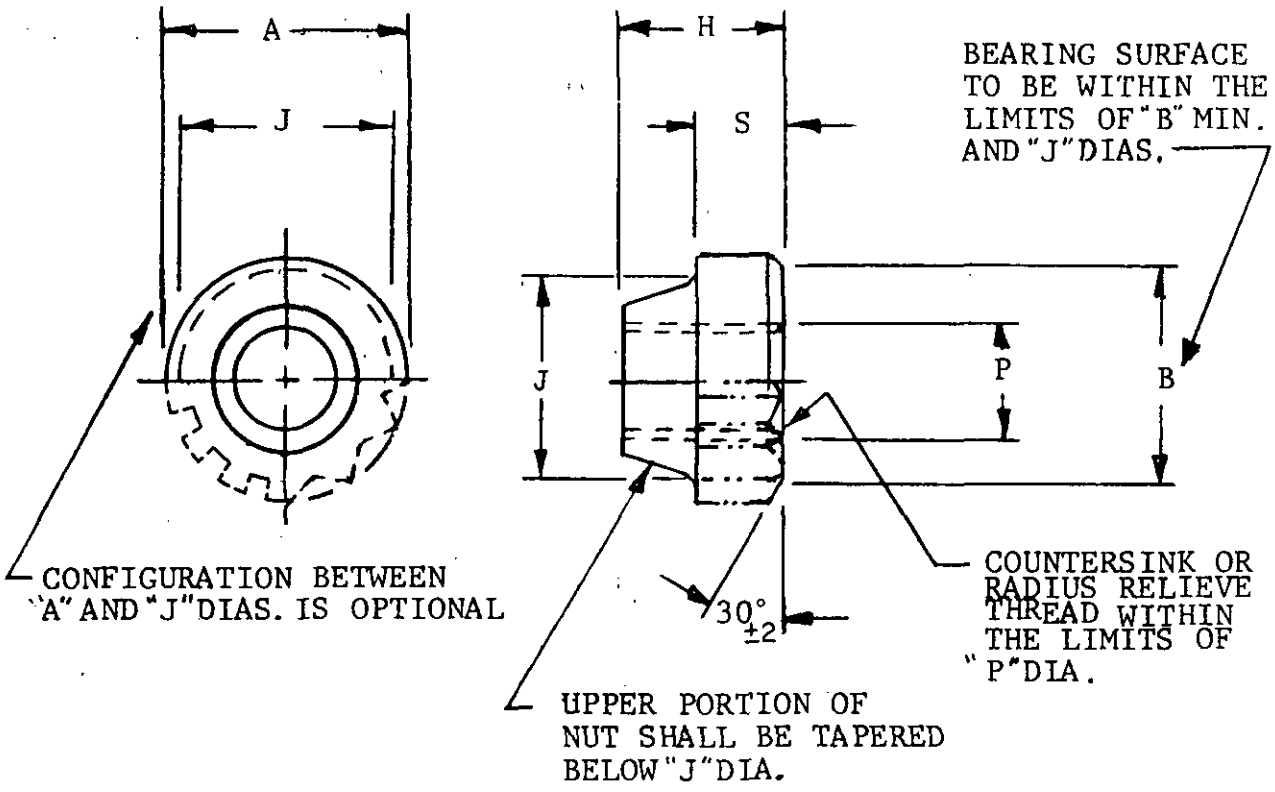


NOTES:

1. Material: Alloy steel.
2. Hardness: Type A - Rc 27-32 for testing bolts rated below 1800MPa.
   Type B - Rc 36-42 for testing bolts rated 1800MPa and higher.
3. Heat treat in accordance with MIL-H-6875.
4. Surface Texture: 3.2 µm in accordance with ANSI B46.1.
5. Protective Coating: Rust preventative oil in accordance with MIL-C-15074 to be removed before testing.
6. Residual traces of cadmium permitted on nuts retapped to remove locking device.
7. Magnetic particle inspect in accordance with MIL-I-6868. Omit identification.
8. Threads in accordance with FED-STD-H28/21.
9. Concentricity: All diameters to be concentric within 0.13 FIM.
10. Squareness: Bearing surface to thread P.D. within 0.08 FIM.
11. Bearing surface to be flat within "Y" FIM but shall not be convex.
12. Unless otherwise specified, dimensions in millimeters.
13. Break all sharp edges.

FIGURE 2. Nut, fatigue test, room temperature, for tension bolts.

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NOTES:

1. Material: Alloy steel.
2. Hardness: Rc 36-42.
3. Heat treat in accordance with MIL-H-6875.
4. Surface Texture: 3.2 µm in accordance with ANSI B46.1.
5. Protective Coating: Rust preventative oil in accordance with MIL-C-15074 to be removed before testing.
6. Residual traces of cadmium permitted on nuts retapped to remove locking device.
7. Magnetic particle inspect in accordance with MIL-I-6868. Omit identification.
8. Threads in accordance with FED-STD-H28/21.
9. Concentricity: All diameters to be concentric within 0.13 FIM.
10. Squareness: Bearing surface to thread P.D. within 0.08 FIM.
11. Bearing surface to be flat within "Y" FIM but shall not be convex.
12. Unless otherwise specified, dimensions in millimeters.
13. Break all sharp edges.

FIGURE 3. Nut, fatigue test, room temperature, for shear bolts.

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TABLE I. Nut dimensions for testing tension bolts.

THREADS METRIC (Note 1)	A DIA MAX	B DIA MIN	H 0 -0.76	J DIA MIN	K DIA MAX	P DIA MAX	Q ±0.25	Y MAX	Y MIN
MJ5 x 0.8	9.02	8.26	6.15	6.40	7.37	6.00	1.47	0.08	0.013
MJ6 x 1	12.22	11.20	8.86	9.65	11.02	7.52	2.08	0.08	0.013
MJ8 x 1	15.40	14.38	10.44	11.30	12.88	9.12	2.56	0.08	0.013
MJ10 x 1.25	18.80	17.78	12.62	12.93	14.73	11.20	3.10	0.08	0.013
MJ12 x 1.25	25.27	24.26	16.89	16.20	18.39	13.87	4.11	0.10	0.025
MJ14 x 1.5	28.68	27.41	18.97	19.48	22.05	15.47	4.67	0.10	0.025
MJ16 x 1.5	31.62	30.35	21.95	21.11	23.88	17.04	5.10	0.10	0.025
MJ18 x 1.5	38.43	37.16	25.53	24.41	27.56	20.22	6.27	0.10	0.025
MJ20 x 1.5	44.96	43.69	29.84	32.64	33.04	21.30	7.31	0.13	0.025
MJ22 x 1.5	44.96	43.69	29.84	32.64	33.04	23.39	7.31	0.13	0.025
MJ24 x 2	51.82	50.29	34.42	36.35	36.73	26.57	8.38	0.13	0.025
MJ27 x 2	58.04	56.52	38.99	37.52	42.24	29.74	9.45	0.15	0.025
MJ30 x 2	64.90	63.37	43.56	42.44	47.73	32.00	10.52	0.18	0.025
MJ33 x 2	72.39	70.10	48.39	45.72	51.38	35.00	11.58	0.18	0.025
MJ36 x 2	78.49	76.20	52.96	49.00	55.04	38.00	12.62	0.20	0.025
MJ39 x 2	85.37	83.36	57.37	52.00	58.70	41.00	13.65	0.20	0.025

NOTES:

1. 4H6H for MJ5 size.
   4H5H for MJ6 size and larger.
2. Dimensions in mm.

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TABLE II. Nut dimensions for testing shear bolts.

THREADS METRIC (Note 1)	A DIA MAX	B DIA MIN	H 0 -0.76	J DIA MIN	P DIA MAX	S REF	Y MAX	Y MIN
MJ5 x 0.8	11.02	9.32	3.83	9.65	6.00	1.45	0.08	0.013
MJ6 x 1	12.88	10.92	5.05	11.30	7.52	2.18	0.08	0.013
MJ8 x 1	14.73	12.50	6.43	12.93	9.12	3.05	0.08	0.013
MJ10 x 1.25	16.53	14.05	7.95	14.53	11.20	4.01	0.08	0.013
MJ12 x 1.25	22.05	18.82	11.15	19.48	13.87	5.59	0.10	0.025
MJ14 x 1.5	25.73	21.97	12.93	22.76	15.47	6.65	0.10	0.025
MJ16 x 1.5	27.56	23.57	14.83	24.41	17.04	7.80	0.10	0.025
MJ18 x 1.5	31.22	26.72	19.05	27.66	20.22	12.83	0.10	0.025
MJ20 x 1.5	36.73	31.47	23.49	32.56	21.30	16.00	0.13	0.025
MJ22 x 1.5	36.73	31.47	23.49	32.56	23.39	16.00	0.13	0.025
MJ24 x 2	42.24	36.25	28.32	37.52	26.57	19.71	0.13	0.025
MJ27 x 2	47.73	41.00	33.25	42.44	29.74	20.62	0.15	0.025
MJ30 x 2	53.24	45.75	38.81	47.37	32.00	21.13	0.18	0.025
MJ33 x 2	58.72	50.50	44.73	52.27	35.00	24.38	0.18	0.025
MJ36 x 2	64.21	55.27	51.05	57.23	38.00	27.86	0.20	0.025
MJ39 x 2	69.70	61.00	58.00	62.44	41.00	30.00	0.20	0.025

NOTES:

1. 4H6H for MJ5 size.
   4H5H for MJ6 size and larger.
2. Dimensions in mm.

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